Status	Finished
Started	Wednesday, 21 October 2026, 10:28 PM
Completed	Wednesday, 21 October 2026, 10:31 PM
Duration	3 mins 6 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Distribution

b. Insertion

c. Extraction

d. Selection

The correct answer is:
Insertion

Question **2**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys

b. A sorting algorithm is stable if it preserves the order of all keys

c. A sorting algorithm is stable if it preserves the order of duplicate keys

d. A sorting algorithm is stable if it preserves the order of non-duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **3**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Bubbling up the smallest element
b. Bubbling up the largest element to its correct position
c. Sorting the entire list in one pass
d. Dividing the list into halves

The correct answer is: Bubbling up the largest element to its correct position

Question **4**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

a. sort()
b. arrange()
c. sorted()
d. order()

The correct answer is: sort()

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Binary Search
b. Merge Sort
c. Linear Search
d. Bubble Sort

The correct answer is: Bubble Sort

Question **6**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By breaking the input into smaller parts and merging them
- b. By sorting data in a single pass
- c. By using the bubble-up method
- d. By comparing elements sequentially

The correct answer is: By breaking the input into smaller parts and merging them

Question **7**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting is rarely used in programming
- b. Sorting makes it harder to search for items
- c. Sorting decreases the efficiency of selection operations
- d. Sorting helps in finding duplicates quickly

The correct answer is: Sorting helps in finding duplicates quickly

Question **8**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Merging
- b. Searching
- c. Complexity
- d. Sorting

The correct answer is:
Complexity

Question **9**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. arrange()
b. sort()
c. sorted()
d. order()

The correct answer is: sorted()

Question **10**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Makes data harder to manage
b. Decreases the efficiency of algorithms
c. Makes code execution slower
d. Provides a basis for other algorithms to work efficiently

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **11**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It increases the number of comparisons needed
b. It has no effect on the search operation
c. It allows for faster searching
d. It makes the search operation slower

The correct answer is: It allows for faster searching

Question **12**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Sorting data sequentially
b. Dividing the input into parts, solving each part, and combining the solutions
c. Rearranging data without sorting
d. Sorting data in a single pass

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **13**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It builds a sorted array one element at a time
b. It compares adjacent elements and swaps them if necessary
c. It repeatedly finds the minimum element and moves it to the sorted part of the list
d. It divides the list into two halves, sorts each half, and then merges them

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **14**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort is less efficient for large lists
b. Bubble Sort does not guarantee sorted order
c. Bubble Sort cannot handle duplicate elements
d. Bubble Sort is difficult to implement

The correct answer is: Bubble Sort is less efficient for large lists

Question **15**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

a. Merging two datasets
b. Finding a specific element in a list
c. Removing duplicates from a list
d. Arranging data in a specific order

The correct answer is: Arranging data in a specific order